Relatório de Análise Exploratória de Dados

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Referência da base de dados

Fonte: Kaggle - Lifestyle and Health Risk

Figure 1: Exemplo de estilo de vida saudável

ANÁLISE EXPLORATÓRIA DE DADOS: PROJETO 04

Carregando os pacotes:

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.4     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(palmerpenguins)
```

```
## Warning: pacote 'palmerpenguins' foi compilado no R versão 4.5.3
```

```
##
## Anexando pacote: 'palmerpenguins'
##
## Os seguintes objetos são mascarados por 'package:datasets':
##
##     penguins, penguins_raw
```

```r
library(ggthemes)
```

```
## Warning: pacote 'ggthemes' foi compilado no R versão 4.5.3
```

```r
library(nycflights13)
library(skimr)
library(readr)
library(gt)
library(corrplot)
```

```
## Warning: pacote 'corrplot' foi compilado no R versão 4.5.3
```

```
## corrplot 0.95 loaded
```

```r
library(GGally)
```

```
## Warning: pacote 'GGally' foi compilado no R versão 4.5.3
```

```r
library(summarytools)
```

```
##
## Anexando pacote: 'summarytools'
##
## O seguinte objeto é mascarado por 'package:tibble':
##
##     view
```

```r
library(gtsummary)
```

```
## Warning: pacote 'gtsummary' foi compilado no R versão 4.5.3
```

```r
library(rstatix)
```

```
## Warning: pacote 'rstatix' foi compilado no R versão 4.5.3

##
## Anexando pacote: 'rstatix'
##
## O seguinte objeto é mascarado por 'package:stats':
##
##     filter
```

```r
library(patchwork)
```

```
## Warning: pacote 'patchwork' foi compilado no R versão 4.5.3
```

```r
library(tidyr)
library(dplyr)
```

Variáveis da base de dados `Lifestyle and Health Risk Prediction: https://www.kaggle.com/datasets/miadul/li`

```r
df <- read.csv("Lifestyle_and_Health_Risk_Prediction_Synthetic_Dataset.csv")

df_num <- df |>
  select(age, bmi, sleep, exercise) #idade,índice corporal, horas de sono, consumo de açúcar
```

Será analisado: idade, saúde (bmi), comportamento (sono + exercícios)

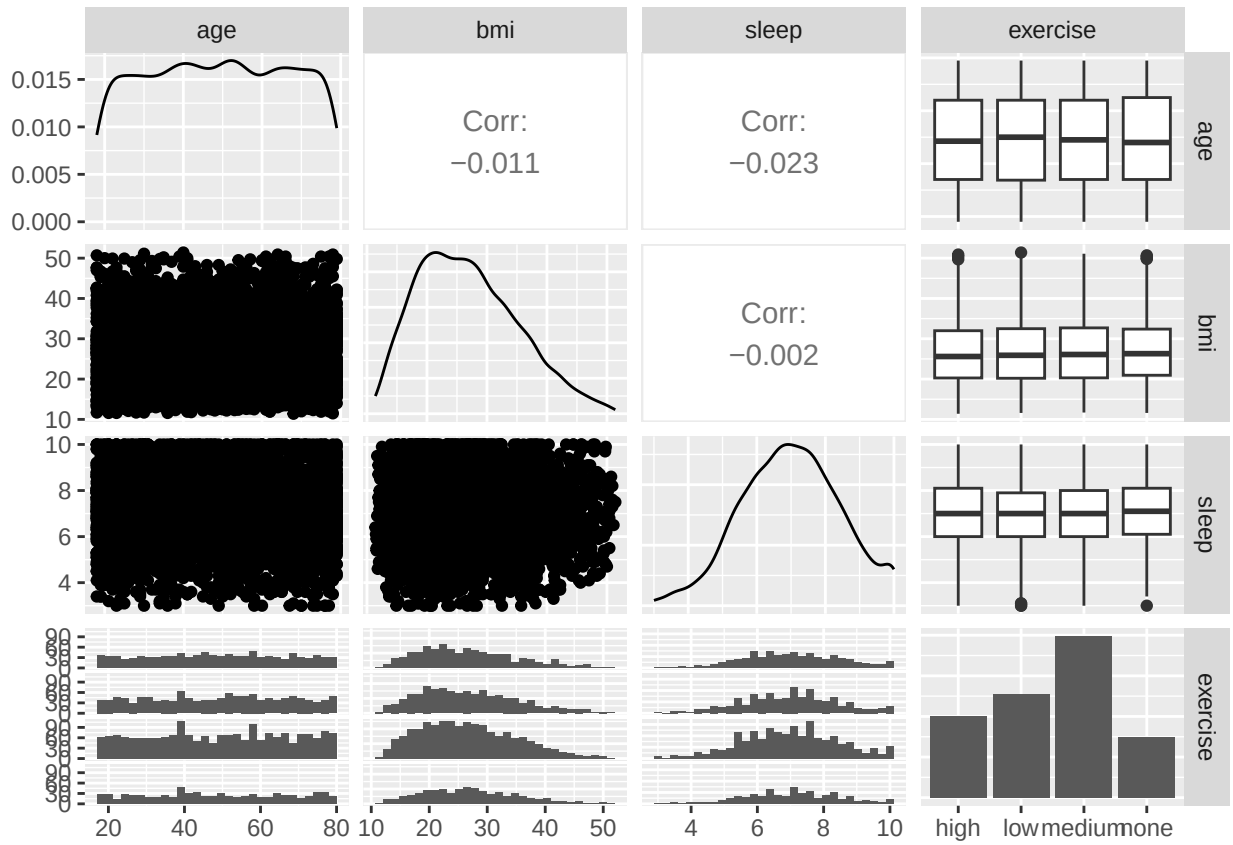
```r
#questão 03
descr(df_num)
```

```
## Non-numerical variable(s) ignored: exercise
```

```
## Descriptive Statistics
## df_num
## N: 5000
##
##                          age       bmi     sleep
## ----------------- --------- --------- ---------
##              Mean     48.81     26.84      7.00
##           Std.Dev     17.91      8.25      1.43
##               Min     18.00     11.40      3.00
##                Q1     34.00     20.30      6.00
##            Median     49.00     26.00      7.00
##                Q3     64.00     32.40      8.00
##               Max     79.00     51.40     10.00
##               MAD     22.24      8.90      1.48
##               IQR     30.00     12.10      2.00
##                CV      0.37      0.31      0.20
##          Skewness     -0.01      0.47     -0.06
##       SE.Skewness      0.03      0.03      0.03
##          Kurtosis     -1.18     -0.36     -0.33
##           N.Valid   5000.00   5000.00   5000.00
##                 N   5000.00   5000.00   5000.00
##         Pct.Valid    100.00    100.00    100.00
```

```r
#questão 04
library(GGally)

ggpairs(df_num)
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



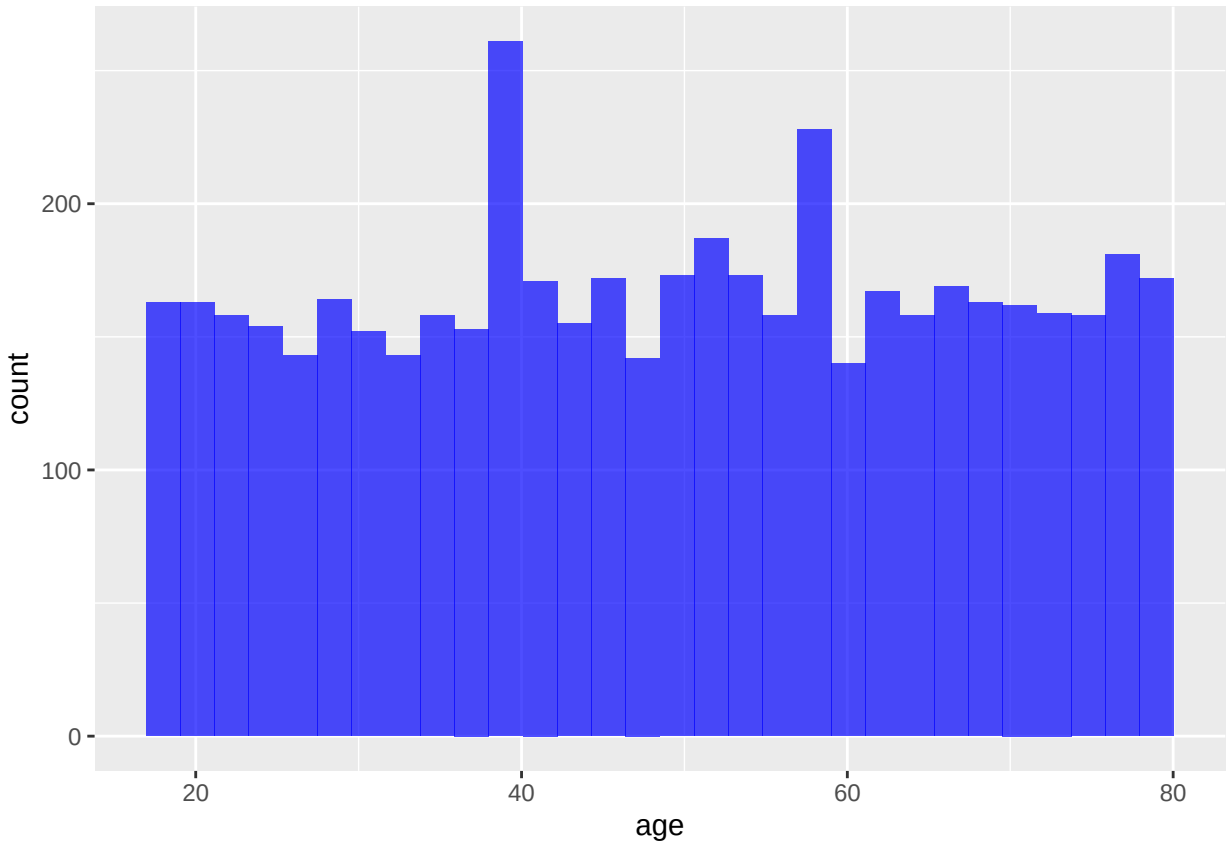

```r
#questão 05


df_num <- df |>
  select(age, bmi, sleep, weight)

#necessário trocar a variável exercise por uma númerica de verdade "weight"
```

```r
str(df_num)
```

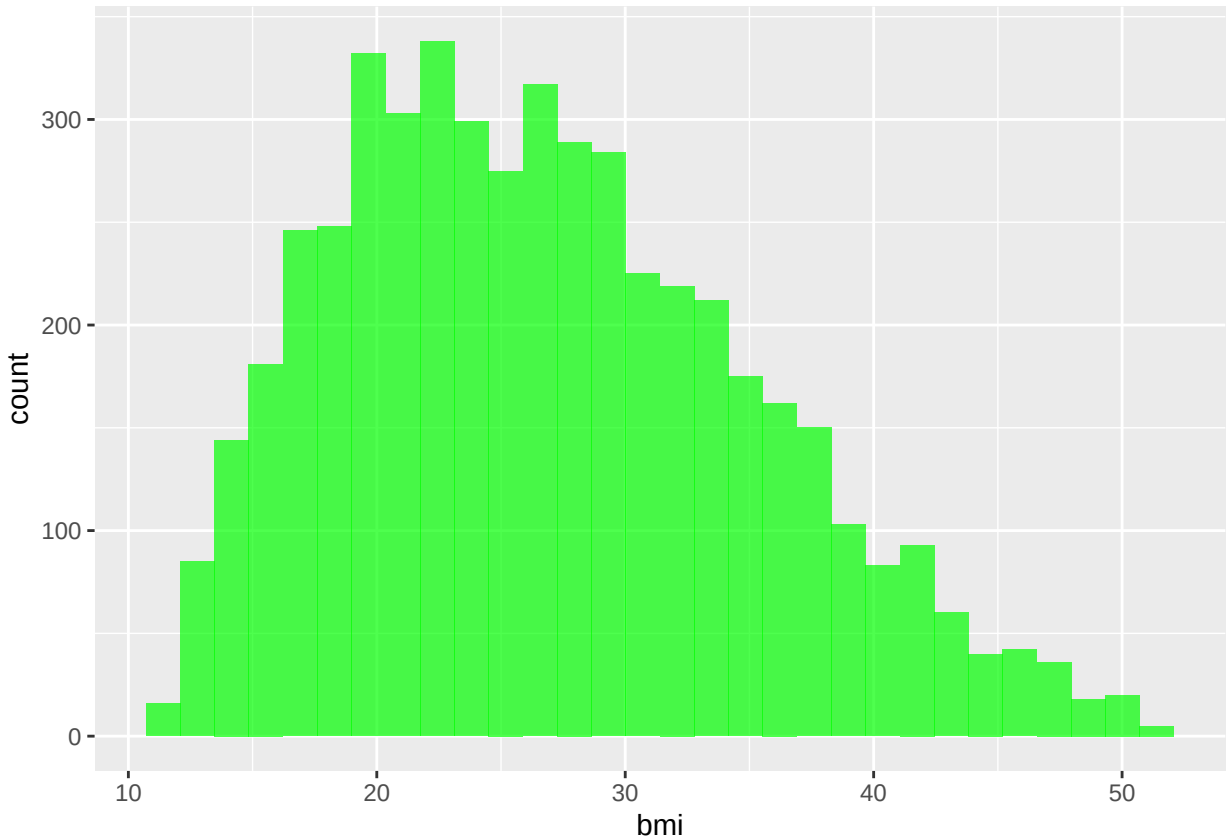
```
## 'data.frame':    5000 obs. of  4 variables:
##  $ age   : int  56 69 46 32 60 25 78 38 56 75 ...
##  $ bmi   : num  17.6 26.3 45.3 15.6 25.8 37.5 22.7 20.2 23.2 43.5 ...
##  $ sleep : num  6.1 6.9 6.6 8.5 8 3.8 9.9 6.6 9.6 8.1 ...
##  $ weight: int  67 76 106 54 98 96 64 76 58 94 ...
```

```r
library(ggplot2)
```

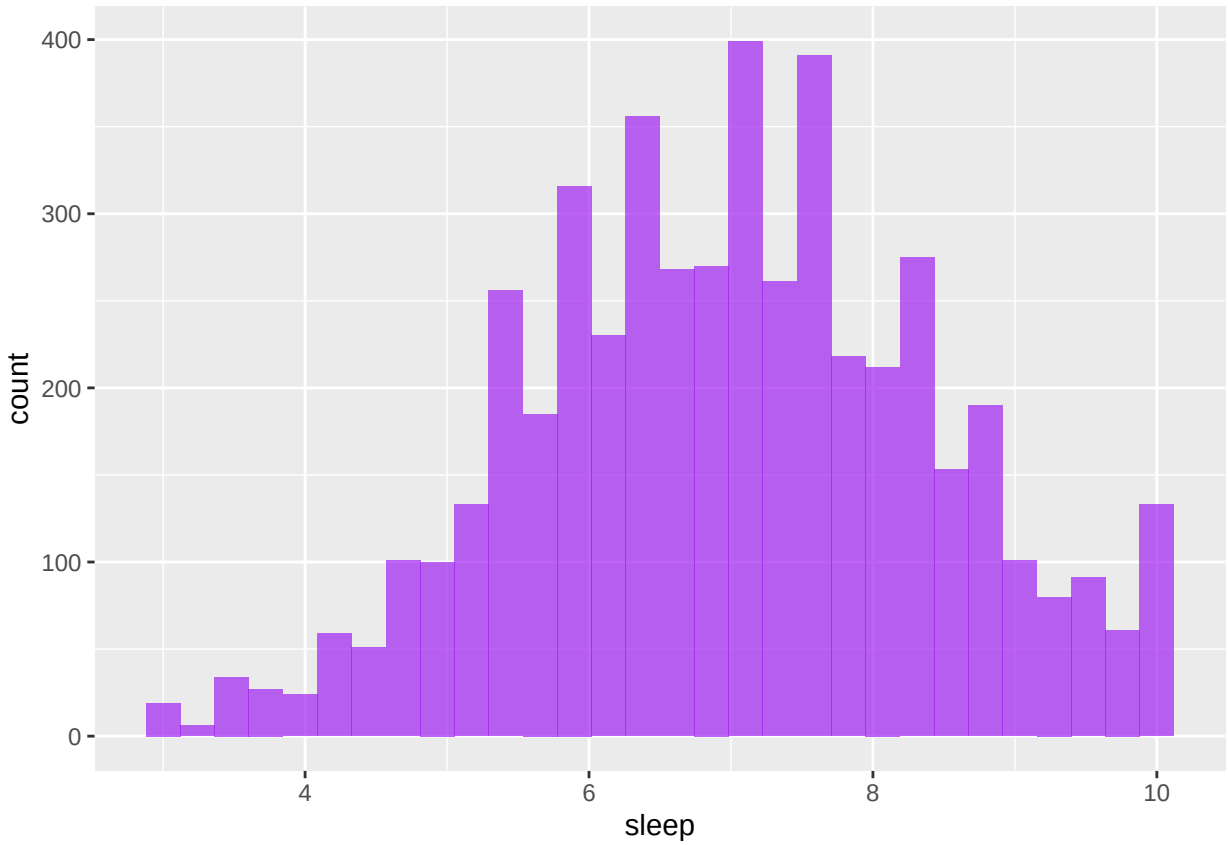
```r
df_num <- df |>
  select(age, bmi, sleep, weight)

# Histograma - age
ggplot(df_num, aes(x = age)) +
  geom_histogram(bins = 30, fill = "blue", alpha = 0.7)
```



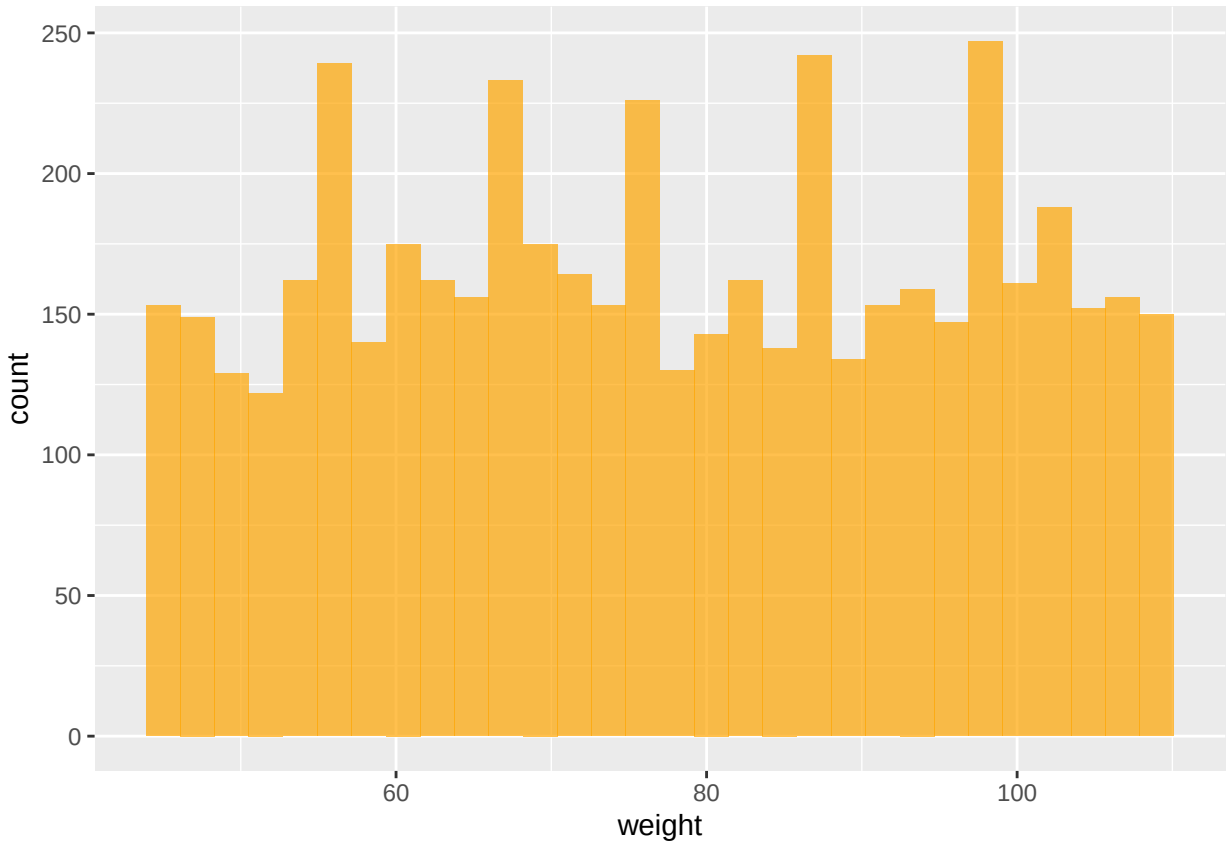

```r
# Histograma - bmi
ggplot(df_num, aes(x = bmi)) +
  geom_histogram(bins = 30, fill = "green", alpha = 0.7)
```




```r
# Histograma - sleep
ggplot(df_num, aes(x = sleep)) +
  geom_histogram(bins = 30, fill = "purple", alpha = 0.7)
```



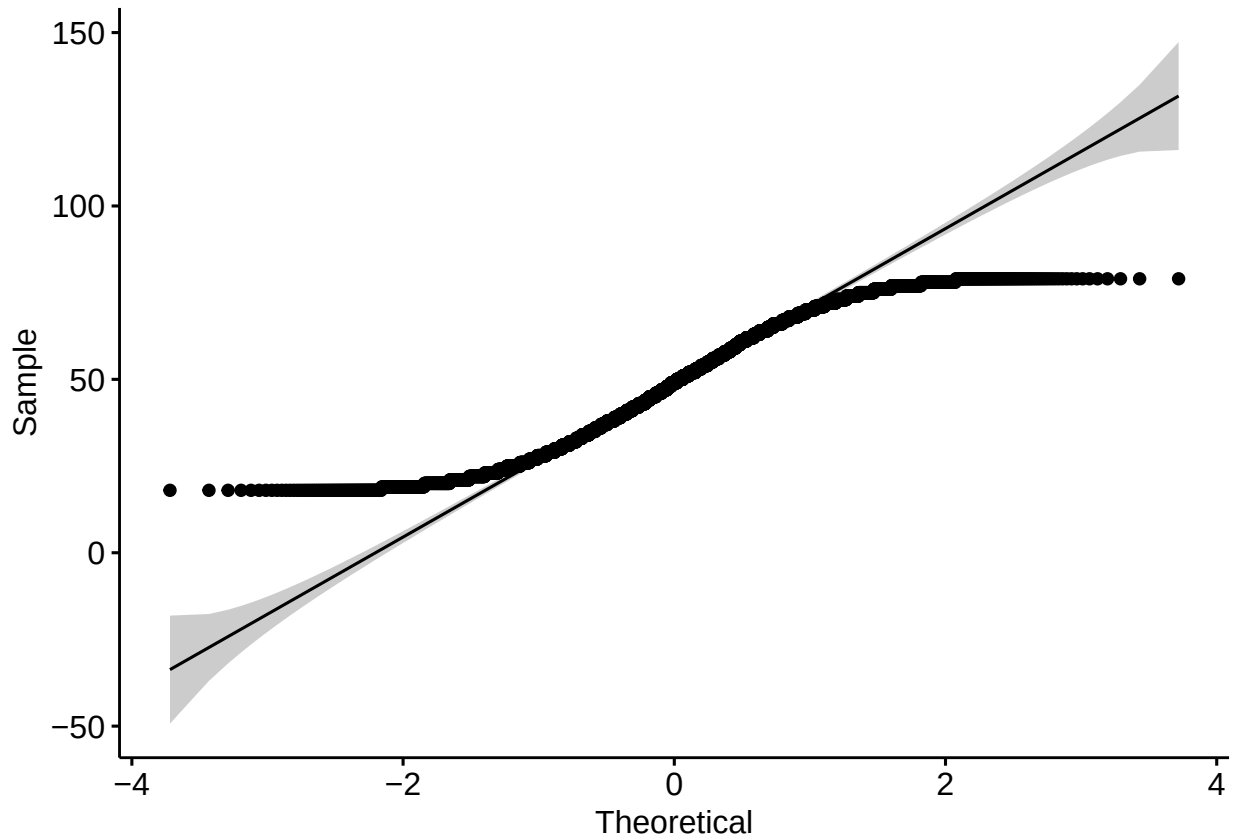

```r
# Histograma - weight
ggplot(df_num, aes(x = weight)) +
  geom_histogram(bins = 30, fill = "orange", alpha = 0.7)
```



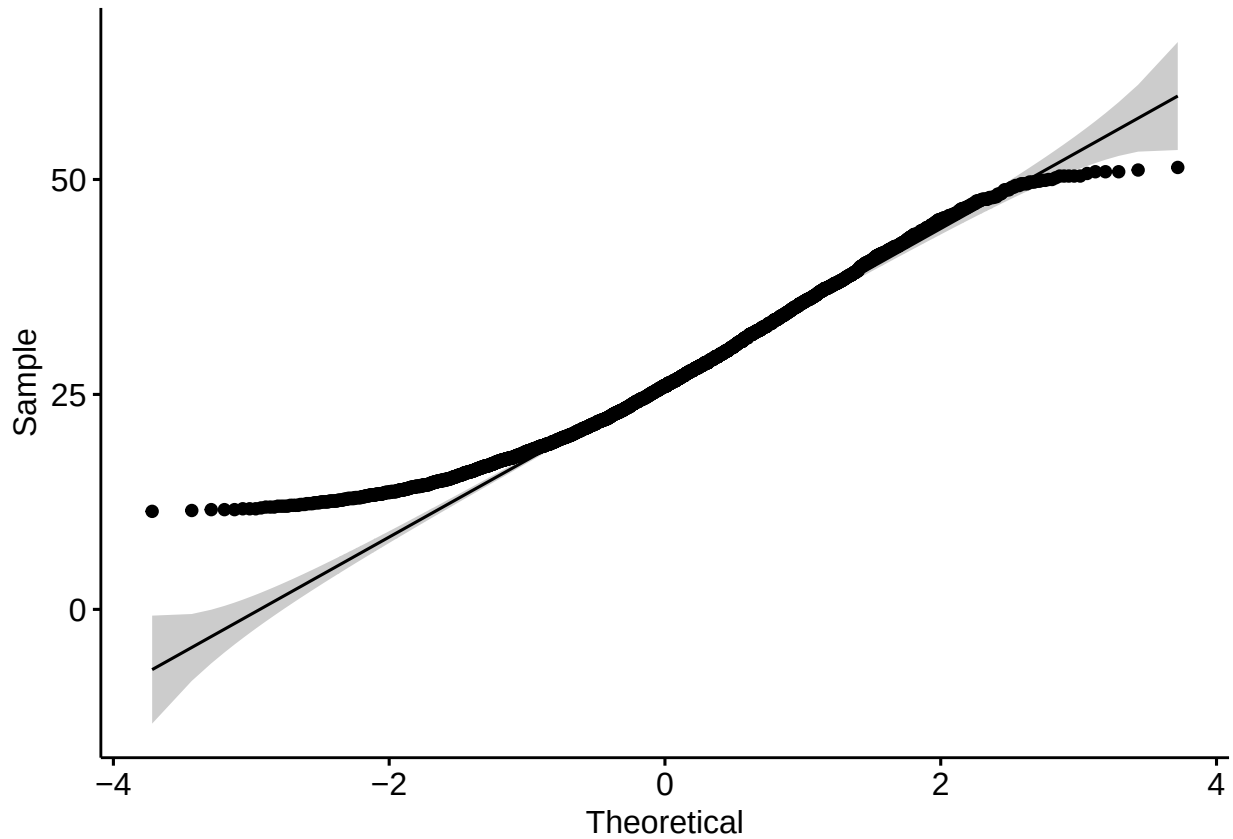

```r
#questão 05 letra c:
library(ggpubr)
```

```
## Warning: pacote 'ggpubr' foi compilado no R versão 4.5.3
```

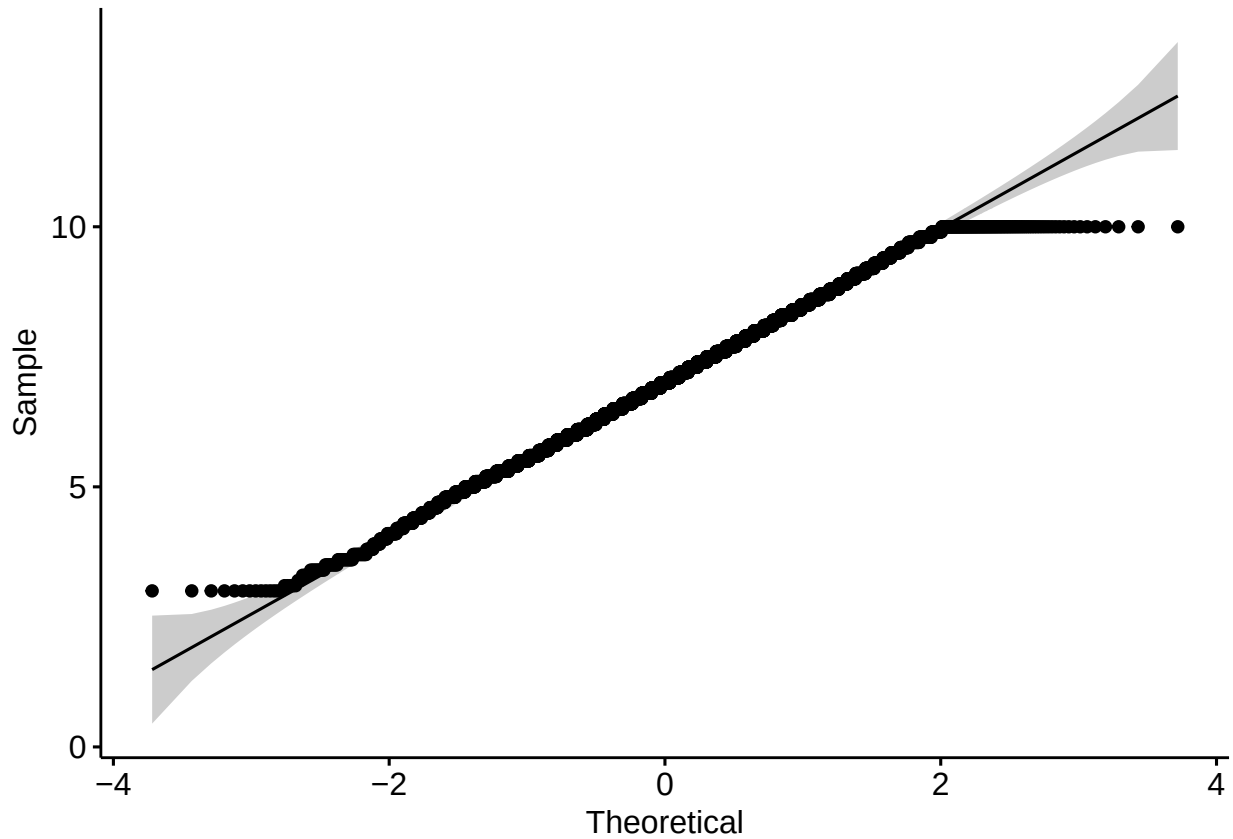
```r
# Q-Q plot - age
ggqqplot(df_num$age)
```



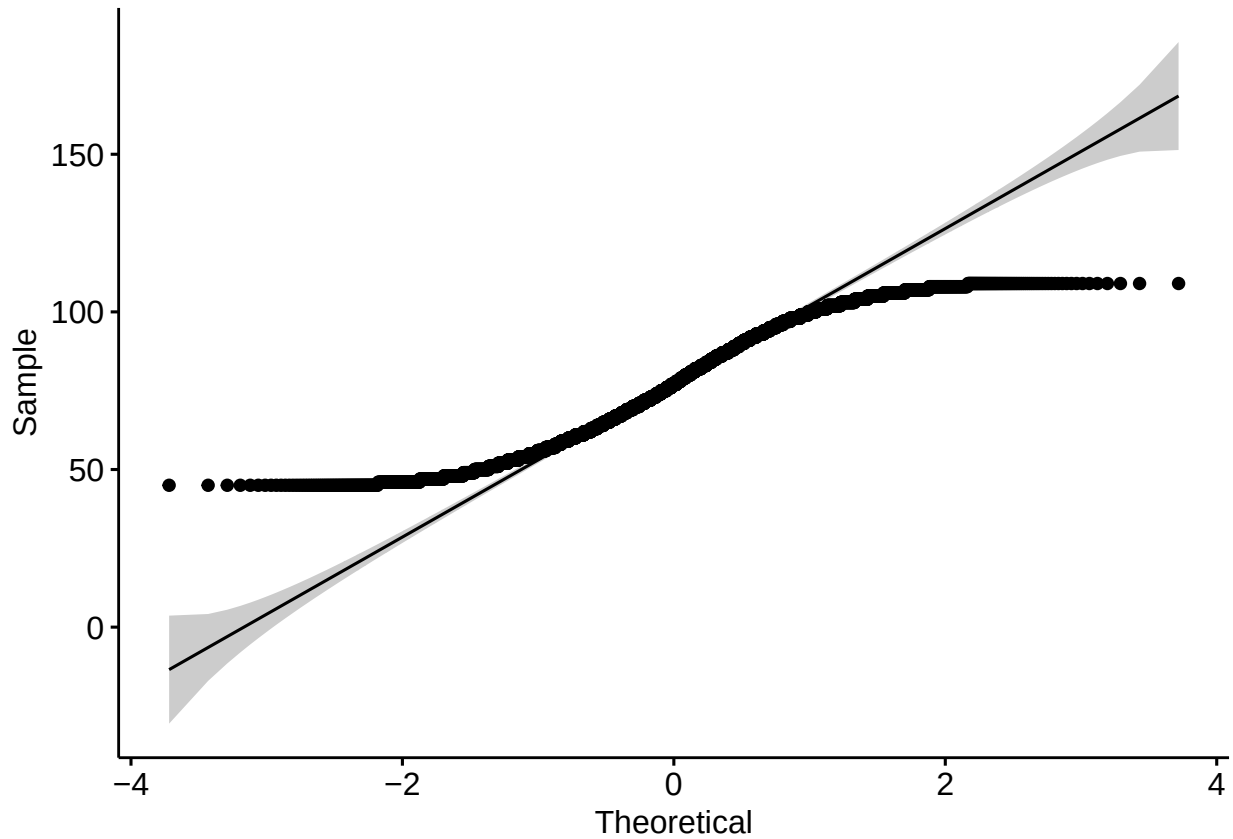

```r
# Q-Q plot - bmi
ggqqplot(df_num$bmi)
```




```r
# Q-Q plot - sleep
ggqqplot(df_num$sleep)
```




```r
# Q-Q plot - weight
ggqqplot(df_num$weight)
```




```r
#questão 5 letra d:
shapiro.test(df_num$age)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_num$age
## W = 0.9553, p-value < 2.2e-16
```

```r
shapiro.test(df_num$bmi)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_num$bmi
## W = 0.97541, p-value < 2.2e-16
```

```r
shapiro.test(df_num$sleep)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_num$sleep
## W = 0.99413, p-value = 1.968e-13
```

```r
shapiro.test(df_num$weight)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_num$weight
## W = 0.95387, p-value < 2.2e-16
```

```r
#questão 07, cálculo da completudo:
# Percentual de valores não nulos (completude)

completude <- sapply(df, function(x) {
  sum(!is.na(x)) / length(x) * 100
})

completude
```

```
##          age       weight       height     exercise        sleep sugar_intake
##          100          100          100          100          100          100
##      smoking      alcohol      married   profession          bmi  health_risk
##          100          100          100          100          100          100
```

Correção para inserir dados faltantes:

```r
set.seed(123)

df_missing <- df

# Inserindo NA em 5% dos dados
df_missing$bmi[sample(1:nrow(df), 0.05 * nrow(df))] <- NA
df_missing$sleep[sample(1:nrow(df), 0.05 * nrow(df))] <- NA
```

```r
#questão 07 editada para dados faltantes:

completude <- sapply(df_missing, function(x) {
  sum(!is.na(x)) / length(x) * 100
})

completude
```

```
##          age       weight       height     exercise        sleep sugar_intake
##          100          100          100          100           95          100
##      smoking      alcohol      married   profession          bmi  health_risk
##          100          100          100          100           95          100
```

```r
#questão 08
library(mice)
```

```
## Warning: pacote 'mice' foi compilado no R versão 4.5.3
```

```
## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car

##
## Anexando pacote: 'mice'

## O seguinte objeto é mascarado por 'package:stats':
##
##     filter

## Os seguintes objetos são mascarados por 'package:base':
##
##     cbind, rbind
```

```r
df_mice <- df_missing |>
  select(age, bmi, sleep, weight)

imputacao <- mice(df_mice, m = 5, method = 'pmm', seed = 123)
```

```
##
##  iter imp variable
##   1   1  bmi  sleep
##   1   2  bmi  sleep
##   1   3  bmi  sleep
##   1   4  bmi  sleep
##   1   5  bmi  sleep
##   2   1  bmi  sleep
##   2   2  bmi  sleep
##   2   3  bmi  sleep
##   2   4  bmi  sleep
##   2   5  bmi  sleep
##   3   1  bmi  sleep
##   3   2  bmi  sleep
##   3   3  bmi  sleep
##   3   4  bmi  sleep
##   3   5  bmi  sleep
##   4   1  bmi  sleep
##   4   2  bmi  sleep
##   4   3  bmi  sleep
##   4   4  bmi  sleep
##   4   5  bmi  sleep
##   5   1  bmi  sleep
##   5   2  bmi  sleep
##   5   3  bmi  sleep
##   5   4  bmi  sleep
##   5   5  bmi  sleep
```

```r
df_imputado <- complete(imputacao)
```

```r
#questão 09
#em um RScript
```